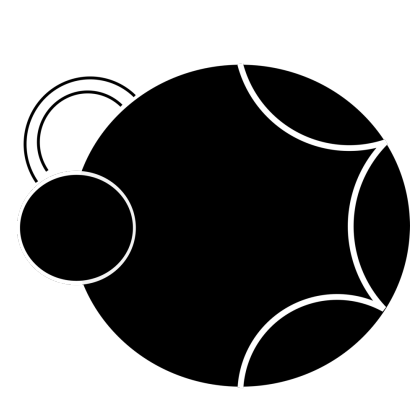


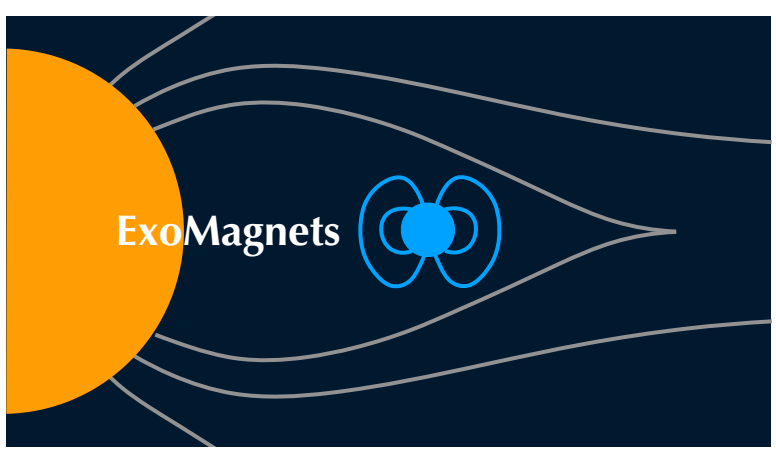


LDE3



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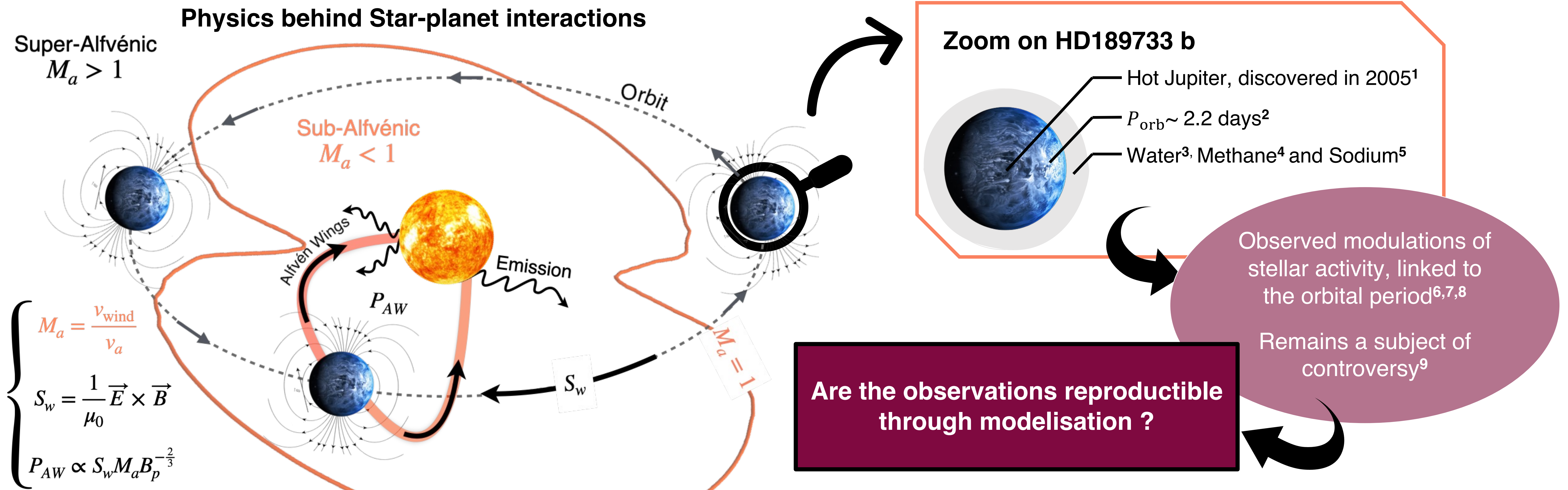
Université Paris Cité

A key hidden factor behind planetary habitability: how to characterise exoplanetary magnetic field?

Context

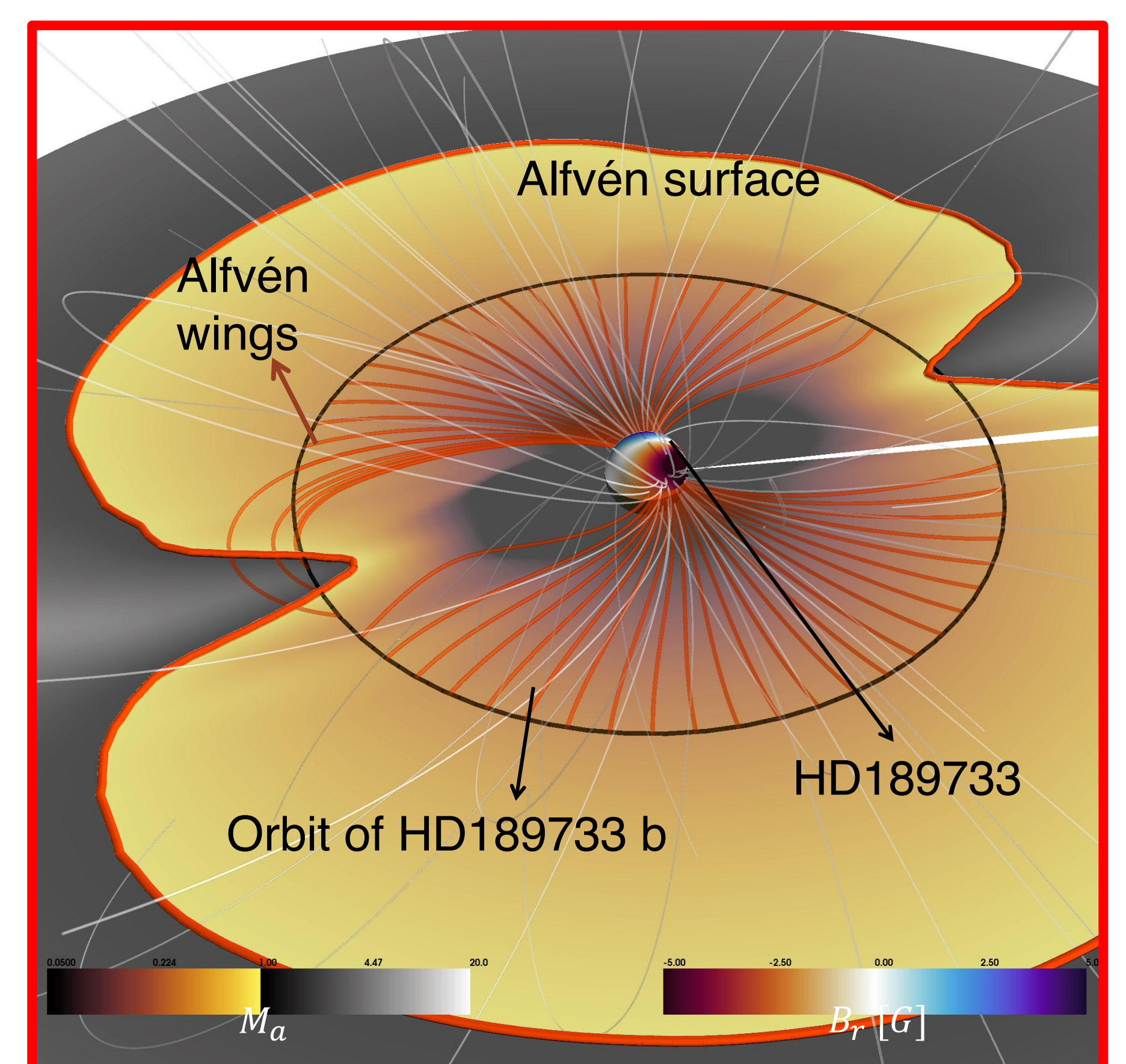
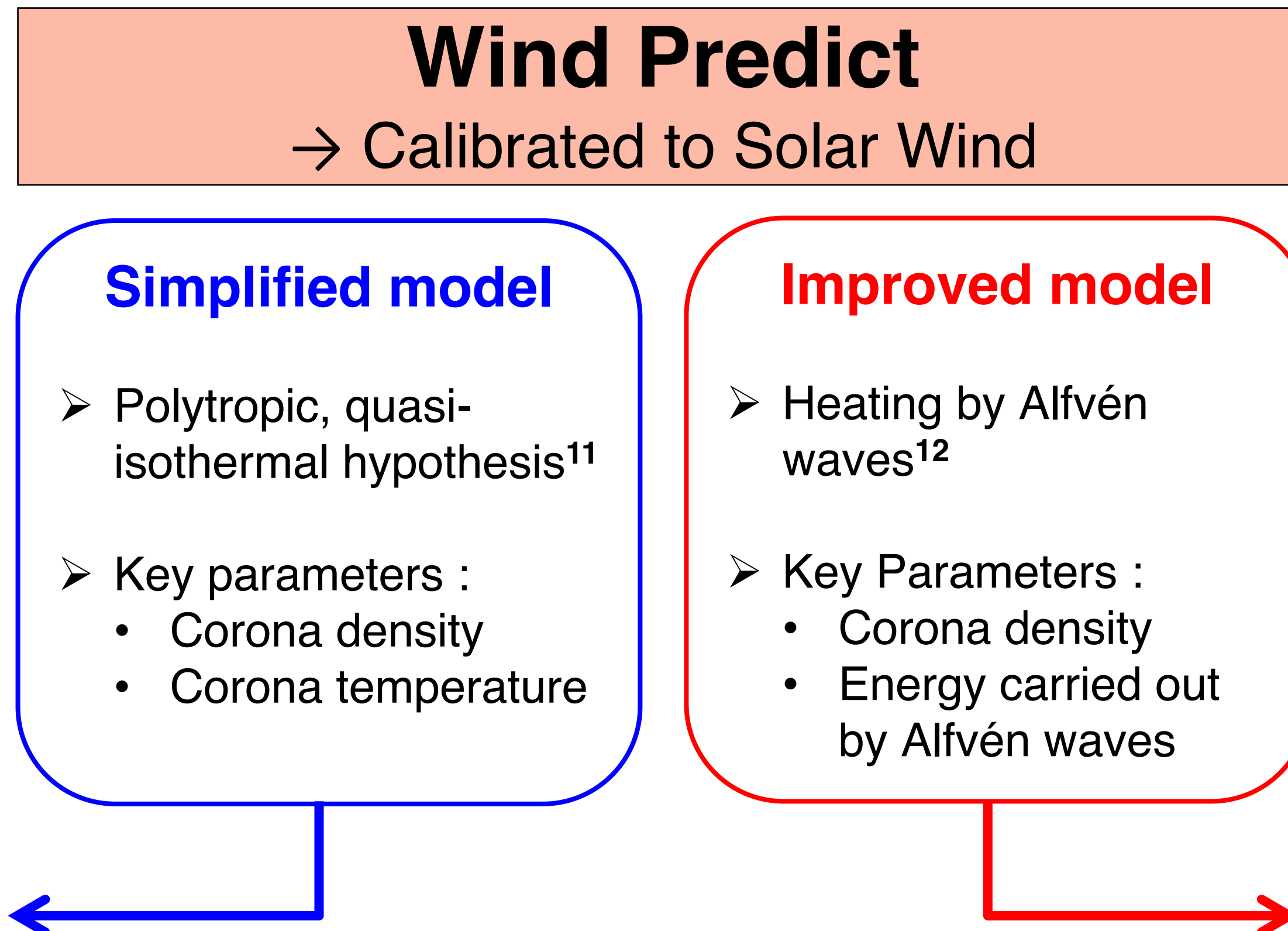
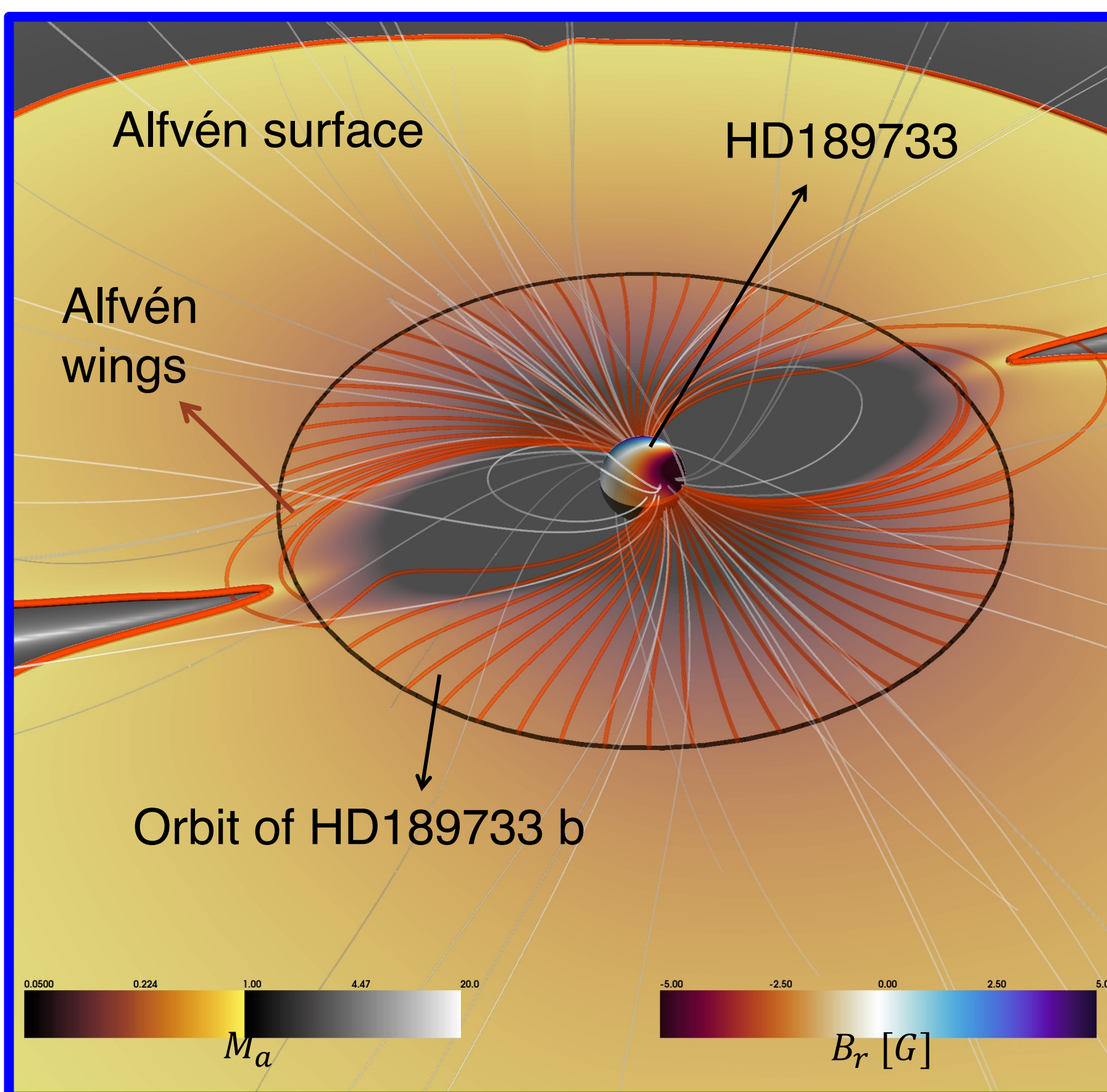
Magnetic field can influence a planet's ability to retain its atmosphere, which is a key factor for habitability (protection against atmospheric erosion and cosmic ray). Yet, not a single exoplanetary magnetic field has been observed at the moment.

➤ Indirect detection of exoplanetary magnetic field by Star-Planet Magnetic Interactions



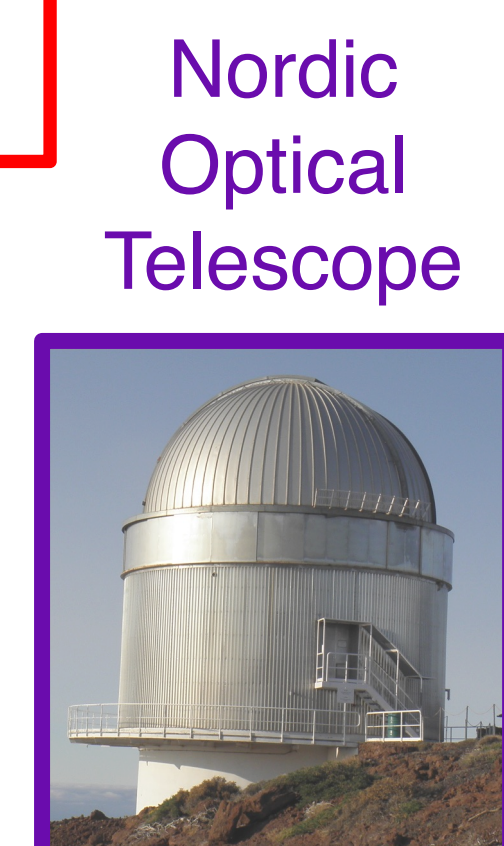
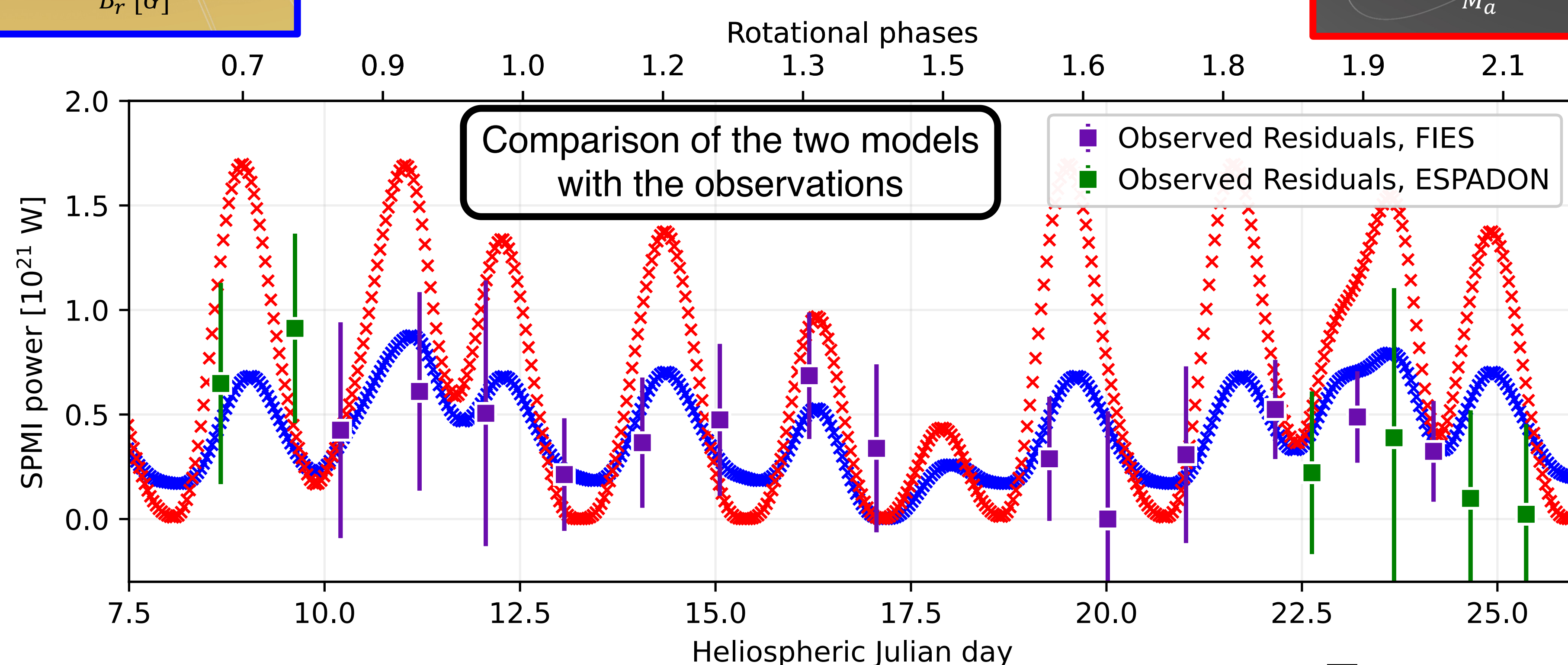
✓ Need to **model** the 3D environment of the star : use of PLUTO¹⁰ to resolve MHD equations in order to characterise **stellar winds**.

✓ Observation of the stellar magnetic field at the surface of the star in Zeeman-Doppler imaging to **constrain the stellar magnetic field**.



Spectro-polarimetric observational campaign led in 2023 in K-band photometry

Strugarek, Moutou & Fares



Nordic Optical Telescope

Telescopes used for the observations of HD 189733 system



Canadian France Hawaii Telescope

Conclusion

- ✓ We reproduced the power generated by star-planet interactions in the HD 189733 system and compared it to the densest observational study ever carried out on HD 189733 star-planet magnetic interactions
- ✓ However, the observational study is still not dense enough to obtain conclusive statistical tests of correlation between the models and the data.

Future Directions

- ❑ Study of other bands of the spectrum (here, K band)
- ❑ Conduct additional observation campaigns with even denser coverage
- ❑ Realistically model an exoplanet by including a magnetosphere, an internal dynamo and an atmosphere in order to better characterise habitability

→ Contact at clemence.gourves@cea.fr

Visually, **simplified model** fits the **data points** better than the **improved model**

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